

FaceKit: a lightweight toolkit for interpretable facial phenotyping in rare diseases

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Abstract

Facial phenotyping plays an important role in the diagnosis and clinical characterization of rare genetic diseases, many of which are associated with recognizable craniofacial features. However, traditional approaches for describing facial morphology rely largely on qualitative clinical observation and free-text descriptions, which are often subjective, non-standardized, and difficult to reproduce across observers and institutions. Although the Human Phenotype Ontology (HPO) provides controlled terms for describing dysmorphic facial features, these terms are typically categorical rather than quantitative and may vary depending on examiner experience and interpretation. Here, we present FaceKit, a computational framework for quantitative facial phenotyping from frontal facial photographs. FaceKit extracts standardized measurements of facial landmarks and derived 120 morphological features, then reports feature-level z-scores representing deviation from population reference distributions. To support ancestry-aware phenotyping, we constructed reference models using the FairFace dataset as a control population spanning diverse ancestry groups. We further evaluated FaceKit using multiple rare disease facial phenotype resources, including GMDB, RFace, and an internal clinical cohort. In addition to quantitative facial analysis, FaceKit includes synthetic facial image generation to support rare disease model development and data augmentation. We also performed privacy evaluation to assess whether synthetic images reveal identifiable information from real patient photographs and could therefore compromise patient privacy. Across disease case studies, FaceKit-derived quantitative measurements captured known facial features associated with rare genetic disorders and provided objective support for clinical phenotype interpretation. We further demonstrate that FaceKit can assist human reviewers in generating HPO terms and text-based descriptions of facial features from patient photographs, improving consistency and reproducibility of facial phenotype annotation. Together, these results establish FaceKit as a useful tool for rare disease research, quantitative dysmorphology, and AI-assisted phenotyping. By converting facial morphology into standardized, quantitative, and interpretable features, FaceKit has the potential to improve rare disease diagnosis, support genotype-phenotype studies, and enable more reproducible clinical characterization across diverse patient populations.

Keywords: facial phenotyping, rare disease, facial photo, landmarking, privacy evaluation

1 Introduction

Facial morphology is an important component of phenotypic characterization in many rare genetic diseases and can provide useful signals for disease recognition alongside other clinical and genetic

information. Traditionally, these patterns have been assessed through expert observation and described using standardized phenotype terminology such as the Human Phenotype Ontology (HPO). More recently, neural-network and vision-language models have also used facial images to support rare-disease recognition and the identification of facial phenotype terms.

However, existing approaches do not fully address the need for quantitative and interpretable facial phenotyping. HPO terms provide a standardized vocabulary, but many facial features are still evaluated qualitatively and are not directly linked to reproducible measurements from images. Similarly, learning-based models commonly produce disease rankings, probabilities, or phenotype labels, while the facial characteristics underlying these outputs are often difficult to express. General-purpose facial landmark tools provide image coordinates, but researchers must still define their own normalization procedures, geometric measurements, feature-selection rules, and output formats. This fragmentation makes it difficult to compare facial morphology consistently across patients, diseases, and study cohorts.

To address this gap, we developed FaceKit, a lightweight and interpretable toolkit that converts facial images into structured quantitative representations of facial morphology. FaceKit integrates facial landmark extraction and visualization, frontal-pose quality filtering, geometric normalization, and cohort average-face generation. It computes 120 interpretable measurements across major facial regions and supports disease-aware feature selection for downstream cohort analysis. FaceKit also provides an extensible framework through which researchers can define additional geometric features while maintaining a unified output representation. Together, these functions provide a reproducible pipeline for transforming facial images into quantitative phenotypic features for rare-disease research.

2 Methods

2.1 Image Preprocessing

FaceKit accepts facial images provided either in a single directory or in a hierarchical directory structure organized by disease cohort. Each image is first processed using MediaPipe Face Landmarker to detect 478 three-dimensional facial landmarks and estimate a facial transformation matrix. The transformation matrix is converted into yaw, pitch, and roll angles to assess whether the image is sufficiently frontal for subsequent geometric analysis. By default, images with an absolute yaw or pitch greater than 15° , or an absolute roll greater than 10° , are marked as non-frontal. This quality-control procedure reduces variation introduced by head pose and identifies images for which facial measurements may be unreliable. For each image, FaceKit stores the landmark coordinates together with the image identifier and dimensions. The extracted data can be saved as individual JSON files or combined into a JSONL file for batch processing and downstream phenotype extraction.

2.2 Geometric Phenotype Extraction

FaceKit converts the extracted facial landmarks into structured quantitative representations of facial morphology. Before feature calculation, landmark coordinates are geometrically normalized to reduce variation caused by image position, orientation, and scale. Facial roll is corrected using eye landmarks, after which the landmark configuration is translated to a common reference position. Scale normalization is performed using anatomically defined reference measurements, including bizygomatic width, facial height, midface height, and mouth width, depending on the facial region and feature being measured.

Using the normalized landmarks, FaceKit calculates 120 interpretable geometric features describing regional and global facial morphology. These features include distances, ratios, angles, thickness measurements, curvature-related measurements, asymmetry indices, and facial proportions across the eyebrows, eyes, nose, philtrum, mouth and lips, chin and jaw, forehead, midface and cheeks, and the overall face. Each feature is defined using explicit landmark combinations and geometric operations, allowing its value to be directly inspected and compared across individuals or cohorts.

For disease-organized datasets, FaceKit supports disease-aware feature selection to retain measurements considered relevant to the corresponding disease while preserving a consistent output structure across samples. User-provided disease names can be normalized to MONDO identifiers, with resolved mappings cached for reuse and candidate terms suggested when an exact match is unavailable. Features not selected for a given disease are represented as missing values rather than removed from the output table. FaceKit also allows users to specify additional landmark-based measurements through an input argument or a JSON configuration file, enabling custom features to be incorporated into the same structured feature representation.

2.3 Average-face generation

FaceKit generates average-face representations to summarize shared facial morphology within an image cohort. Facial images are first aligned to a common landmark configuration to reduce variation in position, orientation, and scale, after which the aligned images are combined into a cohort-level average. Users can specify the output resolution and optionally apply edge smoothing, a transparent background, or blending with a destination face. The resulting average face provides an intuitive visual summary of common facial patterns within a cohort and can facilitate qualitative comparisons across disease groups.

2.4 Synthetic Facial Image Generation

FaceKit additionally supports the generation of synthetic facial images. Two usage modes are provided. Users with a sufficiently large image collection can train a generator on their own cohort and sample from the resulting weights, while users without access to such data can sample directly from generators pre-trained on ten rare-disease cohorts and distributed with FaceKit.

Facial photographs collected for rare-disease research are often retrospective and heterogeneous in acquisition quality, and the images used for training are therefore standardized beforehand. Images that cannot be used for morphological modelling may optionally be excluded at this stage, including those obscured by privacy blocks, those of extremely low quality, and those in which facial features are not clearly visible. Grayscale or colour-degraded photographs are then colorized using DDColor [1], and blurred or low-resolution faces are restored and upsampled using GFPGAN [2]. Finally, FaceKit locates the facial region from the detected landmarks, removes the surrounding background, and resizes the retained facial area to 256×256 pixels, so that all training images share a common facial framing and resolution. This preparation applies to the training corpus only and is not required when sampling from an existing generator.

Generators are trained using StyleGAN3 [3]. A latent vector $z \sim \mathcal{N}(0, I)$ is transformed by a mapping network into an intermediate style vector, which modulates the convolutions of a synthesis network to produce an image, while a discriminator is trained in parallel to distinguish real from generated images; the two networks are optimized adversarially (Fig. 1). Each cohort, comprising on average more than one hundred prepared images, is trained separately at 256×256 resolution with an R1 regularization weight of $\gamma = 2$, horizontal-flip augmentation enabled to increase effective data

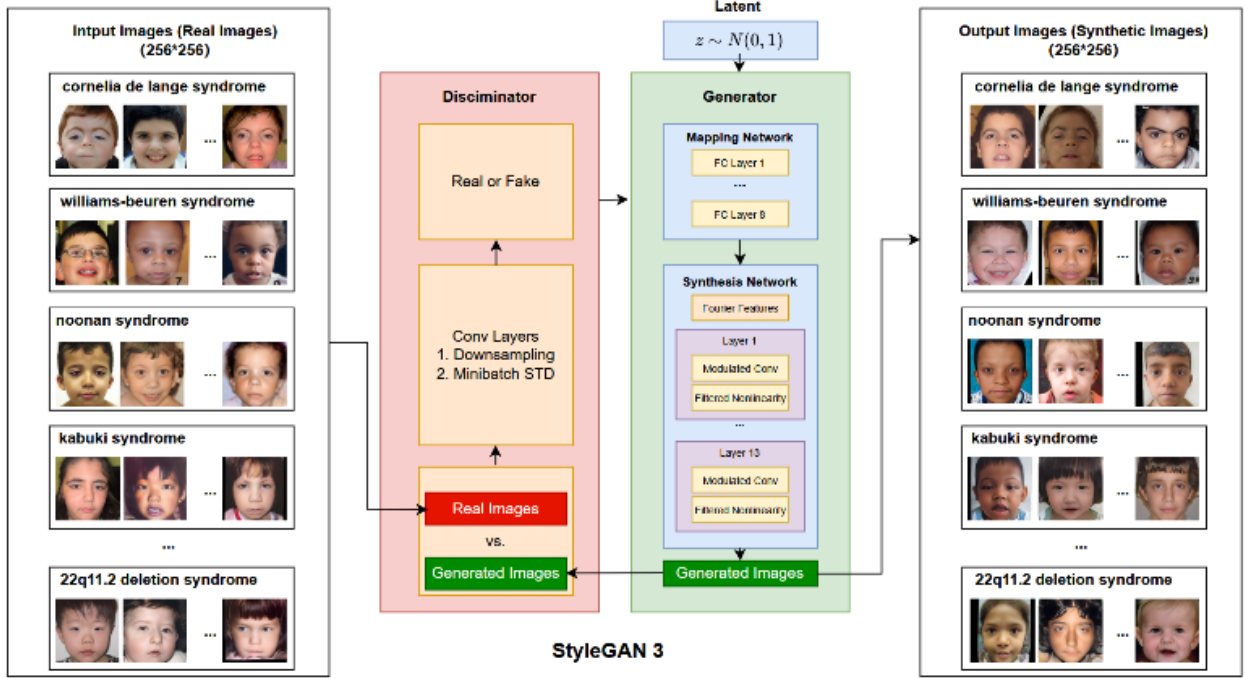


Figure 1: **Synthetic facial image generation.** Prepared real images of each rare-disease cohort (left) are used to train a StyleGAN3 generator. The discriminator applies successive downsampling convolutions with a minibatch standard-deviation layer and separates real from generated images, while the generator maps a latent vector $z \sim \mathcal{N}(0, I)$ through an eight-layer mapping network into a style vector that modulates the convolutions of the synthesis network. Sampling from the trained generator yields 256×256 synthetic facial images for the corresponding cohort (right).

diversity, and a training length of 5,000 kimg. The generator and discriminator learning rates are initialized at 2.5×10^{-3} and 2×10^{-3} , respectively, and follow a cosine decay schedule. Synthetic images are then obtained by drawing latent vectors from the standard normal distribution and passing them through either a user-trained or a pre-trained generator.

2.5 Privacy assessment of the pre-trained generators

Because the generators distributed with FaceKit are trained on photographs of real individuals, the synthetic images they produce are assessed before release for residual similarity to the training data. Two aspects are examined separately: whether a synthetic image reproduces the identity of a training individual, and whether it reproduces the appearance of a training photograph.

For the identity assessment, the prepared images of each cohort are divided into a training partition, which is used to fit the generator, and a held-out partition, which is disjoint from it at the patient level and never enters generator training. Synthetic images are then sampled from the corresponding generator. Faces are detected with InsightFace [4], and the five detected keypoints are used to align each face to a 112×112 crop through a similarity transform. Identity embeddings are extracted with ArcFace [5], and the entire procedure is repeated with AdaFace [6], using an IR-50 backbone trained on MS1MV2, and with LVFace [7], using the LVFace-L model trained on Glint360K, so that the assessment does not depend on a single recognition model; images in which no face is detected are excluded. Distances between embeddings are measured as cosine distances, and every synthetic and every held-out image is characterized by its distance to the nearest training

image of the same cohort. Rather than fixing a verification threshold in advance, the threshold is read off the held-out distribution: for a given percentile p , it is set to the p -th percentile of the held-out nearest-neighbour distances, so that the operating point corresponds to a fixed false-acceptance rate among real images of that cohort. The proportion of synthetic images falling below this threshold is recorded at $p = 1, 2, 5, 10$ and 20 . The ordinal behaviour of each embedding model is verified on the same images: same-person pairs are formed from images of one patient acquired no more than half a year apart, and each such pair is compared against a sample of different-person pairs drawn from the same cohort, checking whether the same-person distance is the smaller one throughout.

The appearance assessment follows the same design with a perceptual rather than an identity metric. LPIPS distances [8] are computed with an AlexNet backbone on images resized to 256×256 , and each synthetic and held-out image is compared exhaustively against every training image of its cohort, retaining the smallest distance; thresholds are again taken from the percentiles of the held-out distribution. Both assessments are finally summarized by the nearest neighbour adversarial accuracy [9]. For two sets of images, the adversarial accuracy is the proportion of images whose nearest neighbour in the other set is farther away than their nearest neighbour within their own set, averaged over the two directions, so that a value of one half indicates that the two sets cannot be distinguished by proximity. It is evaluated between the synthetic images and the training partition, and between the synthetic images and the held-out partition, and the difference between the two, termed the privacy loss, expresses whether synthetic images lie closer to the training data than held-out real images of the same cohorts do. The quantity is computed with embedding distances and with LPIPS distances, pooled across cohorts, and confidence intervals are obtained by resampling images with replacement.

2.6 Landmark visualization

FaceKit provides visualization functions for inspecting facial landmark extraction results on individual images. Detected landmarks can be rendered as a connected facial mesh or as landmark-only points overlaid on the original image. Users can optionally display specific landmark groups, including facial contours and iris landmarks, to examine particular facial regions in greater detail. These visualizations support manual inspection of landmark placement, identification of potential detection failures, and verification of image suitability before downstream geometric phenotype extraction.

3 Results

3.1 Quantitative measurements reproduce annotated facial phenotypes

We applied FaceKit to a curated subset of the GestaltMatcher Database comprising 4,559 frontal facial images from 50 rare-disease cohorts. For each image, FaceKit extracted 478 facial landmarks and computed 120 unitless geometric measurements. After pose filtering and collapsing each patient to the median of their frontal images, 2,897 patients remained. To express every measurement on a common, clinically interpretable scale, we built a normative reference from 2,051 FairFace control images spanning three ancestry groups, of which 886 passed the same pose filter, and converted each measurement into a z -score relative to that reference. Each of the 97 facial HPO terms in the FaceKit vocabulary is mapped to one measurement together with the direction in which that measurement is expected to change, so a signed z -score can be read directly as agreement or disagreement with the annotation.

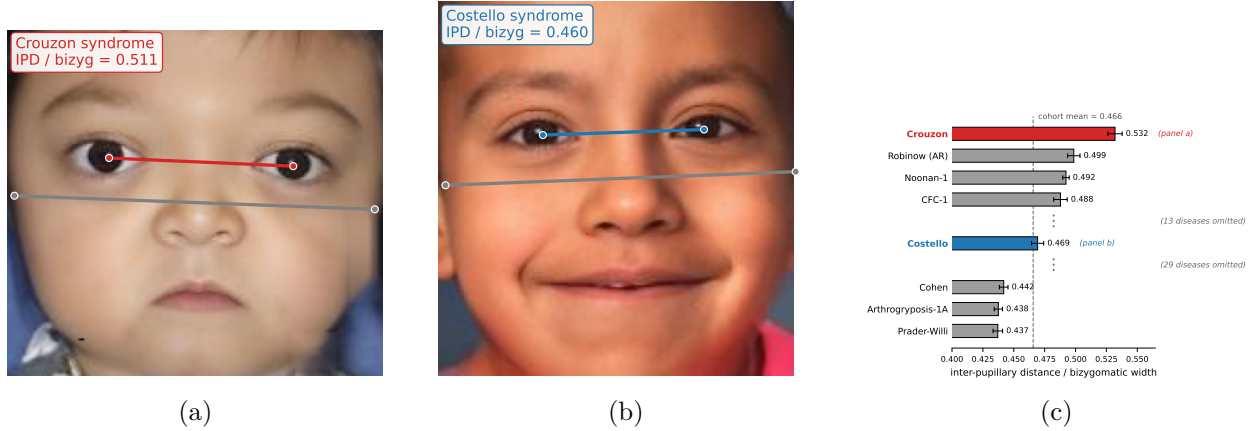


Figure 2: **A single FaceKit measurement, from image to cohort.** (a, b) The inter-pupillary distance normalized by bizygomatic width, drawn on the images it is computed from: the coloured line joins the two iris centres (landmarks 468 and 473) and the grey line joins the two zygomatic points (landmarks 234 and 454); the measurement is the ratio of their lengths. (a) A patient with Crouzon syndrome, the cohort with the highest mean value. (b) A patient with Costello syndrome, a cohort at the centre of the same distribution, shown for contrast. (c) Mean normalized inter-pupillary distance per disease cohort. Bars give the cohort mean with its standard error; the dashed line marks the mean over the remaining cohorts. Of the 50 cohorts, the four highest, the three lowest and Costello syndrome are shown, and the rest are omitted as indicated. Values are computed on images passing the frontal-pose filter.

Of the 97 terms, 84 were carried by at least one patient with a gold-standard annotation; the remaining 13 had no annotated patient anywhere in the cohort and could not be assessed. Across the 84 assessable terms, the mapped measurement deviated from the healthy reference in the direction predicted by HPO for 62 (73.8%; binomial $p = 1.5 \times 10^{-5}$), with a mean absolute deviation of 0.45 reference standard deviations. Restricting attention to the 78 terms carried by enough patients to admit a statistical test, 32 reached significance after Benjamini–Hochberg correction, 42 did not separate from the reference, and 4 deviated significantly in the direction opposite to the one HPO predicts. Agreement was not driven by the terms with the most annotation: terms carried by 10–29 patients agreed as often as those carried by 30 or more (82.4% versus 79.5%).

Hypertelorism illustrates how an individual measurement can be read. The term is mapped to the inter-pupillary distance normalized by bizygomatic width, the ratio of the distance between the two iris centres to the distance between the two zygomatic points (Fig. 2a,b). Across the 238 patients annotated with the term, this ratio was elevated by 0.94 reference standard deviations ($q = 3.9 \times 10^{-20}$), and among all 120 measurements it was the third most deviant in these same patients, indicating that the signal is specific to the measurement the term is mapped to rather than a general distortion of the face. The same ordering is visible at the cohort level: mean normalized inter-pupillary distance was 0.532 in the Crouzon syndrome cohort against 0.466 across the remaining cohorts, the highest of the 50 diseases (Fig. 2c). Panels a and b contrast a Crouzon patient with a patient from a cohort at the centre of the same distribution, with the two landmark pairs entering the ratio drawn on the image.

The four measurements that moved against their annotation share a single explanation, and it is not that the annotations are wrong. Deviations in this paragraph are reported as raw measurement z -scores, positive meaning a larger measurement, rather than oriented to the direction the term

predicts. In each case the annotated patients were barely distinguishable from the patient cohort as a whole: patients annotated with a narrow nasal bridge measured +0.82 against a cohort-wide baseline of +0.52, facial asymmetry -0.67 against -0.49 , and full cheeks -0.60 against -0.51 . For synophrys the annotated patients were in fact closer to the reference than the average patient (+0.39 against +0.49). These are therefore not phenotype effects with an inverted sign but cohort-wide offsets in the measurement itself, which happen to point away from the direction the term predicts.

Each offset has an identifiable source. Every asymmetry measurement, not only the one mapped to facial asymmetry, was systematically lower in patients than in controls (14 of 14 such columns negative, mean -0.30): both the canonical-mesh prior of the landmark detector and the de-rotation step regularize a face toward bilateral symmetry, so the quantity the term depends on is suppressed before it can be measured. Cheek area is normalized by the square of bizygomatic width, and the patients annotated with full cheeks did have broader and rounder faces (face aspect ratio +0.44, roundness +0.29 against cohort baselines of +0.16 and +0.10); the denominator therefore grows with the phenotype and inverts the ratio, the same failure mode that normalization by a phenotype-sensitive scale produces elsewhere. For synophrys, the measurement is the gap between the medial brow endpoints, but synophrys is hair occupying that gap, and the landmark mesh carries no photometric information: the brow points are placed by the shape prior rather than by the hair boundary. Consistent with this, the brows of these patients did differ from the reference in shape, but in a different measurement (medial flare -0.72) than the one the term is mapped to. Only the narrow nasal bridge result rests on few patients ($n = 19$), and it uses landmark indices that have not been verified against the canonical face model.

These four cases indicate that a measurement should be interpreted against the patient population as well as against healthy controls, since a cohort-wide offset can produce a confident result in either direction, and that normalization scales must be chosen so that the denominator is not itself altered by the phenotype being measured.

3.2 Synthetic images reproduce cohort-level geometric phenotypes

We next asked whether the synthetic faces produced by the trained generators carry the same geometric phenotype signal as the real photographs they were trained to imitate. For ten of the rare-disease cohorts we compared 1,727 synthetic images against the 1,647 real images of the same syndromes, passing both sets through the identical FaceKit pipeline and comparing them on the same 120 measurements. For each syndrome the comparison was restricted to the measurements its HPO annotations map to, giving 265 syndrome-measurement pairs in total.

Within each syndrome, the synthetic and real distributions of a measurement were close. The median absolute Cohen’s d between the two sets ranged from 0.052 (Rubinstein–Taybi syndrome) to 0.139 (Angelman syndrome), with an overall median of 0.082; 86.0% of all pairs fell below $|d| = 0.2$ and 98.9% below $|d| = 0.5$ (Fig. 3b). A generator therefore reproduces not only the appearance of a cohort but the location and spread of the individual measurements taken from it.

A more demanding test is whether the synthetic images preserve what distinguishes one syndrome from the others, rather than only reproducing each cohort in isolation. For every syndrome we computed the effect of the focal cohort against the nine remaining cohorts pooled, separately in the synthetic and in the real images, and compared the two sets of effects. The direction of the effect agreed between synthetic and real for a median of 87.2% of a syndrome’s measurements, and the effect sizes themselves tracked closely (median Spearman $\rho = 0.887$, median Pearson $r = 0.924$; Fig. 3a). Agreement was weakest for KBG syndrome (71.4% of 35 measurements) and complete for Angelman syndrome, although the latter is mapped to only three measurements.

Across all 265 pairs, 39 (14.7%) disagreed in sign (Fig. 3a, highlighted). These disagreements were concentrated in the eyebrow region (8 of 30 pairs, 26.7%) and the eye region (21 of 126, 16.7%), and were absent from the philtrum and forehead. They were also almost entirely confined to measurements that barely separate the cohorts in the real images to begin with: the median $|d|$ measured in real images was 0.057 for the disagreeing pairs against 0.234 for the agreeing ones, and 97% of the disagreements occurred where the real effect was below $|d| = 0.2$. Where a measurement carries no discriminative signal in the real cohort, its sign is not determined, so these cases indicate an absence of signal to reproduce rather than a signal reproduced incorrectly. The eyebrow region is the exception worth naming: KBG syndrome, the weakest cohort overall, disagrees most on inter-eyebrow distance ($d = +0.225$ synthetic against $d = -0.121$ real), the same measurement that failed for synophrys above, and for the same reason — the phenotype lives in hair rather than in shape, and neither the landmark mesh nor a generator trained through it represents it.

The one systematic difference between synthetic and real images is dispersion. The variance of a measurement in the synthetic set was consistently smaller than in the real set, with a median ratio of 0.83 across all pairs and regional medians from 0.76 in the chin and jaw, eye and eyebrow regions to 0.96 in the philtrum (Fig. 3c). Synthetic cohorts are therefore narrower than the cohorts they were trained on: the generators recover where a syndrome sits in measurement space and how it differs from other syndromes, but under-represent the variation within it. This should be taken into account when synthetic images are used to augment training data, since a model trained on them will see a narrower range of facial morphology than the real population presents.

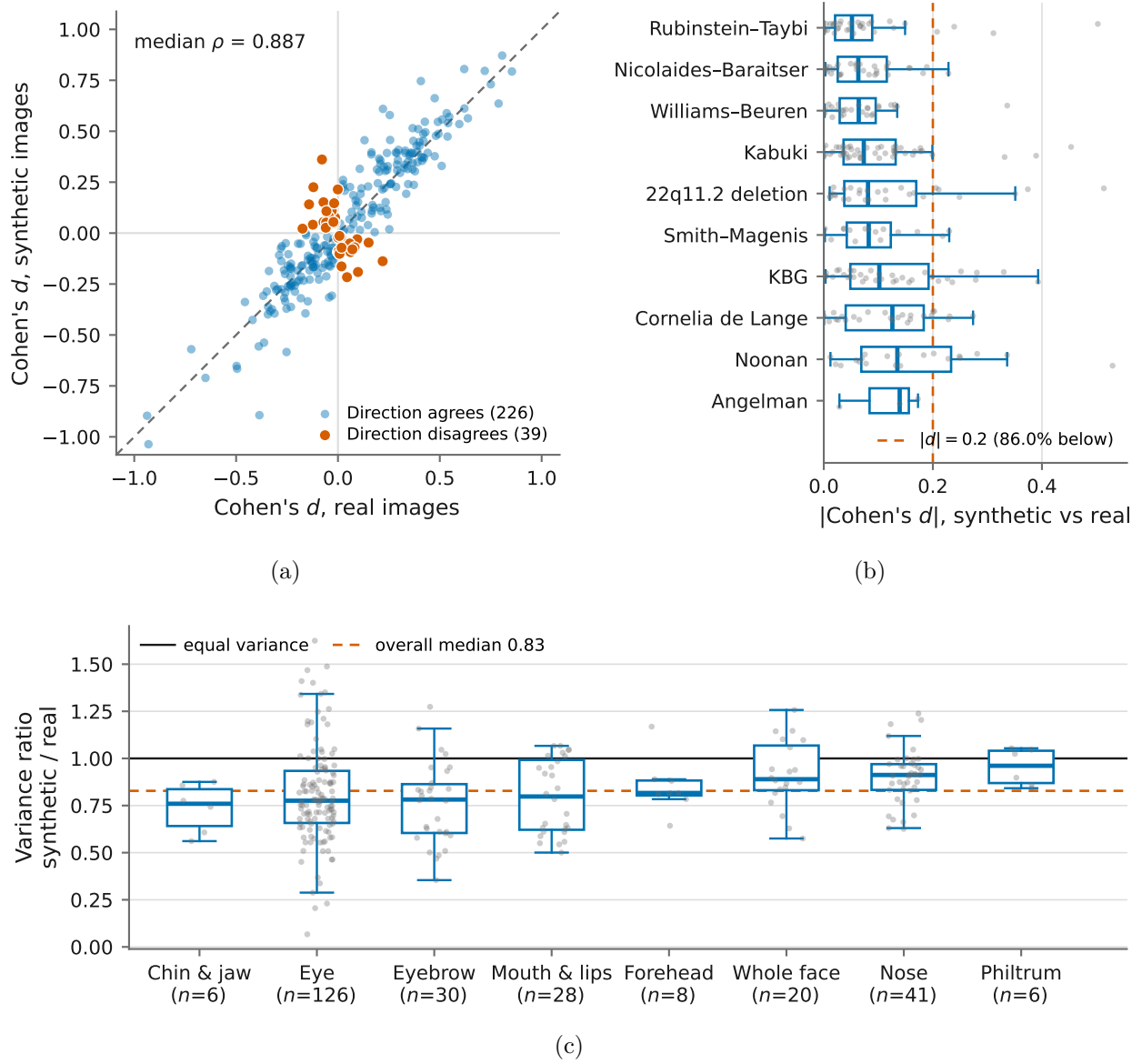


Figure 3: Synthetic cohorts reproduce the geometric phenotype of the cohorts they were trained on. All panels cover the 265 syndrome–measurement pairs obtained by restricting each of the ten syndromes to the measurements its HPO annotations map to. (a) Effect of the focal cohort against the nine remaining cohorts pooled, measured in the synthetic images against the same effect measured in the real images; the dashed line is equality. Pairs whose effect disagrees in sign between the two are highlighted, and lie almost entirely near the origin, where the real effect is itself close to zero. (b) Within each syndrome, the absolute Cohen's d between the synthetic and the real distribution of a measurement. Boxes give the median and interquartile range, whiskers extend to $1.5 \times$ the interquartile range, and individual pairs are drawn behind them. (c) Ratio of the variance of a measurement in the synthetic images to its variance in the real images, by facial region and ordered by median. Values below one indicate a synthetic cohort narrower than the real cohort.

3.3 Synthetic images do not reproduce training identities

Because the generators are distributed with FaceKit, we finally asked whether the images they produce disclose the identity of the patients whose photographs trained them. For the ten rare-disease cohorts we compared 999 sampled synthetic images against the training partition of each cohort, using as a reference the 611 held-out real images that belong to the same cohorts but were excluded from generator training. Face recognition embeddings separate these cohorts less cleanly than they separate adults in the benchmarks they were calibrated on: across 93 same-person and 111,598 different-person pairs, the two ArcFace distance distributions were largely distinct but retained a tail overlap of 0.0716 (Fig. 4a), so no threshold separates identities without error. We therefore did not fix a verification threshold. Each threshold was instead read off the held-out distances themselves, so that an operating point is defined by the false acceptance rate it produces among real images of the same cohort, and the synthetic images were asked only to stay at or below that rate. A held-out image is subject to the same recognition errors and to the same coincidental resemblance between children who share a syndrome as a synthetic image is, so what remains above this reference is the excess attributable to the generator. The comparison requires the embedding to rank identities correctly rather than to be calibrated, and it does: each same-person pair contributes two anchor orderings, and in 182 of these 186 comparisons (97.8%) the distance of the same-person pair was smaller than every different-person distance it was compared against.

Against this reference the synthetic images carried no excess identity signal (Table 1, Fig. 4c). At an operating point corresponding to a 1% false acceptance rate among real images, 0.10% of synthetic images fell below the ArcFace threshold, 0.10% below the AdaFace threshold and none below the LVFace threshold. The margin persisted as the operating point was relaxed: at 5% the three rates were 1.30%, 0.70% and 1.50%, and at 20% they were 15.02%, 11.61% and 6.71%. No backbone exceeded its reference rate at any of the five operating points, and the ordering of the three models was stable throughout. Within the resolution of these three recognition models, and at every operating point examined, sampling from the released generators is therefore not more likely to produce a face matching a training patient than drawing another real photograph of the same syndrome is.

The picture on the appearance axis is different, and the two must not be conflated. LPIPS distances between images of one patient and between images of different patients are almost indistinguishable, overlapping by 0.8438 across 897 same-person and 78,388 different-person pairs (Fig. 4b); the metric responds to texture, pose and illumination rather than to identity. Measured with it, the synthetic images were systematically closer to the training photographs than the held-out images were, and by a wide margin: 14.1% of synthetic images fell below the threshold set at a 5% reference rate, and 36.1% below the threshold set at 20% (Fig. 4d). A generator trained on a cohort reproduces the acquisition characteristics of that cohort, and the synthetic images inherit them. Because the same metric cannot tell two photographs of one patient from photographs of two different patients, this proximity describes the distribution the images were drawn from and not the individuals within it.

Both axes are summarized by the nearest neighbour adversarial accuracy. Computed on embeddings, the privacy loss was -0.000 for ArcFace, -0.002 for AdaFace and -0.006 for LVFace, with bootstrap confidence intervals that included zero in all three cases; synthetic images lie no closer to the training partition than to the held-out partition. Computed on LPIPS distances the privacy loss was $+0.046$ with a confidence interval of $[0.031, 0.093]$ that excludes zero, reproducing in a single number the divergence between the two axes. One feature of these values deserves comment: the adversarial accuracies themselves ranged from 0.947 to 0.980 on embeddings, far above the value of one half that would indicate two indistinguishable sets, because synthetic faces

Table 1: **Synthetic images flagged against a real-image reference rate.** Each row fixes an operating point by taking the threshold from the p -th percentile of the distances between held-out real images and the training partition, so that p is the false acceptance rate the threshold produces among real images of the same cohort; entries give the percentage of the 999 synthetic images falling below that threshold. Thresholds are cosine distances for the three recognition models and LPIPS distances for the perceptual metric. Values at or below p indicate no excess over the reference.

p (%)	Identity (% flagged)			Threshold			Appearance
	ArcFace	AdaFace	LVFace	ArcFace	AdaFace	LVFace	LPIPS (% flagged)
1	0.10	0.10	0.00	0.3758	0.3708	0.5146	3.8
2	0.50	0.20	0.00	0.4115	0.3879	0.5450	7.7
5	1.30	0.70	1.50	0.4386	0.4411	0.5946	14.1
10	5.01	4.00	2.60	0.4750	0.5027	0.6249	23.0
20	15.02	11.61	6.71	0.5274	0.5504	0.6673	36.1

are globally separable from real ones in recognition embedding space. The absolute level therefore carries no privacy interpretation here, and only the difference between the two comparisons does.

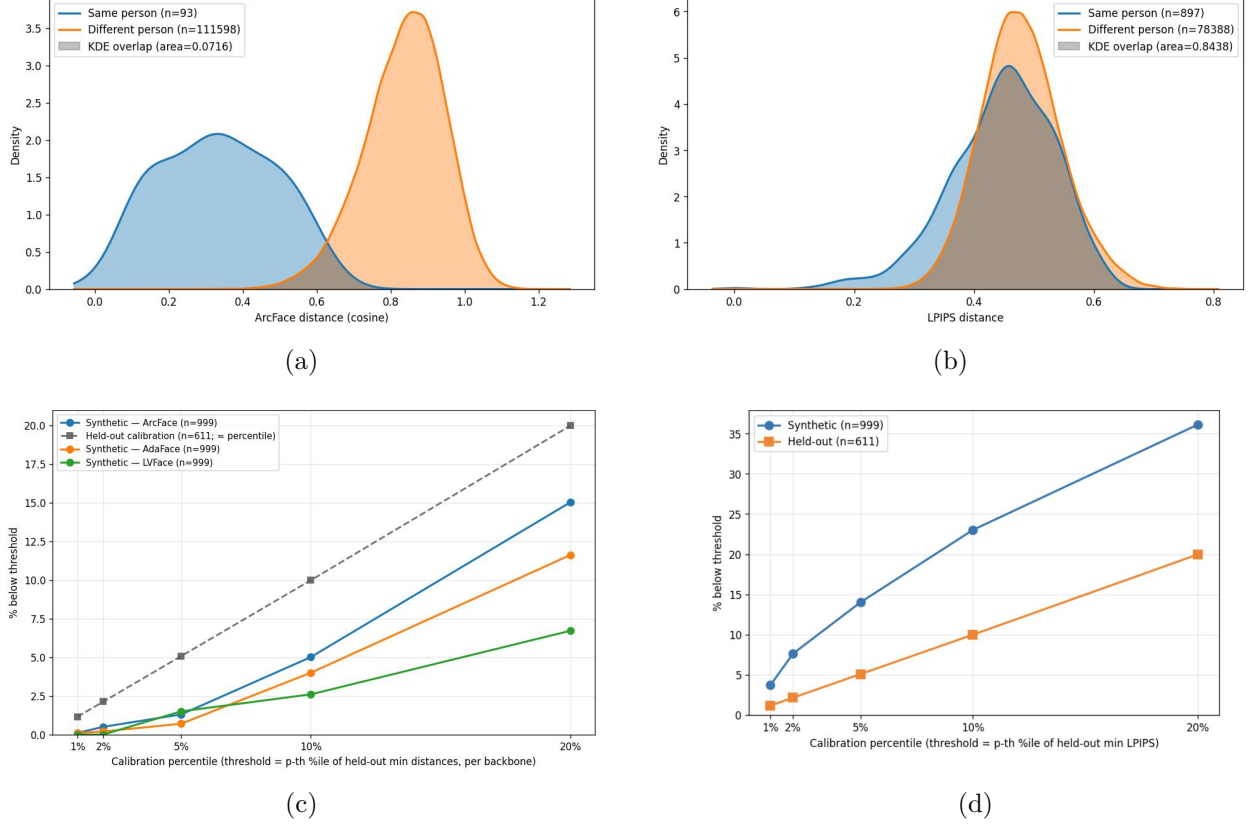


Figure 4: **Privacy assessment of the released generators.** (a) Distribution of ArcFace cosine distances between images of one patient and between images of different patients, with the overlap between the two densities shaded. (b) The same distributions measured with LPIPS, which overlap almost completely and therefore carry little identity information. (c) Percentage of synthetic images falling below a threshold set at the p -th percentile of the held-out distances, for the three recognition models; the dashed line is the rate the same threshold produces among real held-out images. (d) The same comparison measured with LPIPS, where the synthetic images are flagged well above the held-out reference at every operating point.

4 Discussion

FaceKit provides a reproducible and interpretable workflow for transforming facial images into structured geometric phenotype representations. The phenotype-faithfulness analysis showed that most of the mapped measurements deviated from a normative reference in the direction encoded by the corresponding HPO term, although only a subset of these deviations reached statistical significance. At the same time, the results demonstrate that the feature catalogue should be interpreted as a partial geometric representation of facial phenotype rather than a complete translation of HPO descriptions. Individual measurements can be inspected through their landmark definitions and used in cohort-level statistical analyses, but FaceKit is not intended to determine the presence of an HPO term or provide a disease diagnosis.

The privacy assessment supports distributing the pre-trained generators, but with a distinction that should be preserved when the synthetic images are used. On the identity axis, sampling from a released generator was no more likely to produce a face close to a training patient than drawing another real photograph of the same syndrome was, and this held for all three recognition models at

every operating point examined. On the appearance axis the synthetic images were clearly closer to the training photographs than held-out real images were. This reflects the acquisition characteristics the generator learned alongside facial morphology rather than the identity of any individual, since the perceptual metric cannot separate two photographs of one patient from photographs of two different patients. The assessment is nevertheless an empirical audit against specific recognition models and a specific perceptual metric, not a formal privacy guarantee: it bounds what these particular measurements can detect, and a stronger attack or a better embedding could in principle detect more.

Several limitations should be considered. First, the reliability of every geometric measurement depends on the accuracy of the underlying facial landmark detector. General-purpose detectors are primarily trained on normative faces and may place landmarks according to learned expectations of typical facial structure when applied to rare-disease faces with substantially different morphology. Errors at this stage propagate directly into all downstream measurements and cannot be corrected through feature design alone. The same limitation applies to any phenotype defined by appearance rather than by shape, as the synophrys result above illustrates: a landmark mesh carries no photometric information and cannot represent hair distribution. Second, some feature failures arise from the choice of geometric formula or normalization scale, as in the full-cheeks result, indicating that normalization strategies may need to be selected separately for different measurements. Third, landmarks estimated from a single two-dimensional image remain sensitive to head pose, facial expression, image resolution, cropping, and acquisition conditions, despite the pose-quality filtering and geometric normalization implemented in FaceKit. These measurements should therefore not be interpreted as equivalent to calibrated three-dimensional facial measurements.

Future development will focus on supporting landmark-detection models adapted to rare-disease faces, refining feature-specific normalization strategies, and incorporating complementary appearance-based representations for phenotypes that cannot be described by geometry alone. Validation in larger cohorts and across demographic and imaging subgroups will also be needed before quantitative ranges can be interpreted clinically.

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